DIRECTED CONTAINERS ARE THE OBJECT-LEVEL MODEL OF A COMPOSABLE AGENT
A CONVERGENCE OF FOUR PROGRAMMES ON ONE EQUIVALENCE

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ABSTRACT. A directed container is exactly a small category. We argue that this single equivalence — Ahman–Uustalu's **DCont** $\cong$ **Cat** — is the correct object-level model of a *composable agent*, and we support the argument by exhibiting four independent lines of work that all reach for it. The equivalence is a *classification target*: we have proved that the polynomial monad liftings of the Reader monad are, one level down, small categories, and that the sound liftings of the State monad embed one. It is *cited infrastructure*: an external programme in compositional active inference (Smithe, Topos Institute / VERSES) invokes it as settled background — verbatim, "Famously, $\triangleleft$-comonoids correspond to categories" — to build a substrate of open dynamical systems. It is a *generalization frontier*: pullback-preserving comonads on **Set** are known to correspond to Grothendieck toposes with enough points, a topos-theoretic refinement approached from the opposite end. And it is a *charted boundary*: two explicit witnesses (a bag functor and a multigraph comonad) sit just outside the polynomial world and pin down where the equivalence stops. We emphasise one honest distinction. The external programme and ours agree on the *object* — a composable agent is a directed container — but compose agents by structurally different mechanisms: Smithe wires Bayesian lenses in a monoidal bicategory, whereas we weld two agent categories by a Zappa–Szép product whose obstruction is a degree-two cohomology class. Convergence on the object; divergence on the composition law. This is the object-level foundation on which the Kodamai grant's compositional-agent narrative rests.

Provenance grades. We tag each claim. [PROVED]: an author's result, established in a proof session. [LEAN]: machine-checked in the author's Lean development, `sorry`-free. [COMPUTED]: a worked example or computation. [EXTERNAL]: another author's result, cited from a deep-read source. [SIGNPOST]: a real but not-yet-load-bearing pointer (source not deep-read or untracked); named, not leaned on.

1. Introduction

The object-level question. Before one can ask how two agents *compose*, one must answer a prior question: what *is* an agent, as a mathematical object? The applied-category-theory literature on multi-agent systems has several answers — an agent is a sheaf section, a coalgebra, a lens, a node in a wiring diagram — and the profusion is a symptom. Composition theorems are only as good as the objects they compose. If the object is wrong, the composition law inherits the wrongness.

This note makes and defends one answer. *At the object level, a composable agent is a small category — equivalently, a directed container.* An agent exposes an interface: a set S of things it can be asked (requests, prompts, states) and, for each, a set $P(s)$ of things it can reply. That is a container $(S \triangleleft P)$. The agent is moreover *directed*: replies can be composed and there is a null reply, so the container carries the extra structure that makes its **Set**-endofunctor a comonad. Ahman and Uustalu proved that a directed container is exactly a small category [1]. The agent's roles are objects, its admissible reply-chains are morphisms, and its handoff structure is composition. *The object is a category, and the category is a directed container.*

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The claim, and why it is not obvious. That a single interface *is* a small category is a clean statement, but on its own it is just a re-description. What elevates $\mathbf{DCont} \cong \mathbf{Cat}$ from a re-description to a foundation is that it is *load-bearing in four independent ways at once.* During a single week of the author's programme, the same theorem surfaced as the target of a classification, the cited background of an external application, the special case of a topos-theoretic generalization, and the interior of a boundary result. Four activities that look unrelated are one theorem seen from four sides. That is the empirical content of "this is the right object": the best objects are the ones many independent lines of work cannot avoid.

We state the main result of the note — which is a thesis supported by evidence, not a new theorem — on this page.

> *The equivalence* $\mathbf{DCont} \cong \mathbf{Cat}$ *is the object-level model of a composable agent. It is simultaneously (1) a classification target for the monad liftings that model effectful agents, (2) the cited infrastructure of an independent programme in compositional active inference, (3) a special case of a topos-theoretic generalization, and (4) a theorem whose boundary is charted by explicit non-examples. Independent programmes converge on this object; they diverge only on the law by which such objects compose.*

The surprise, and the discipline it demands. The sharpest of the four fronts is the second. Toby Smithe's *Compositional Active Inference II* [4], written from the Topos Institute (later VERSES) and rooted in the free-energy / predictive-coding tradition — a world with no obvious contact with directed containers — reaches for *exactly* our equivalence, and treats it as textbook background. At Proposition 2.7 the paper reads, verbatim, "Famously, $\triangleleft$-comonoids correspond to categories and their comonoid homomorphisms are cofunctors" [4, Prop. 2.7], citing Ahman–Uustalu. Two programmes that began at opposite ends — ours from container combinatorics, Smithe's from Bayesian inference — arrive at the same object.

This convergence is genuine, and it is also easy to overclaim, so we impose a discipline on ourselves throughout. Smithe uses $\mathbf{DCont} \cong \mathbf{Cat}$ as the substrate for open *dynamical systems* (the codiscrete comonoid Sy^S on a state space), not as the mechanism by which agents compose; in his framework agents compose as *Bayesian lenses* in a monoidal bicategory. Our own composition law is different: a Zappa–Szép weld $\mathcal{C} \bowtie \mathcal{D}$ of two agent categories — the directed-container weld of Ahman–Uustalu [2], which we read into agents and equip with an H^2 obstruction. So the honest claim is that independent programmes converge on *the object*, $\mathbf{DCont} \cong \mathbf{Cat}$; it is *not* that anyone but us composes agents by the weld. Section 4 states this contrast precisely, because getting it wrong would turn a real convergence into a false scoop.

Outline. Section 2 recalls the container/category dictionary and fixes the agent-as-container reading. Section 3 presents the four fronts: the classification into the equivalence (3.1), its use as external infrastructure (3.2), its generalization upward (3.3), and its boundary from outside (3.4). Section 4 isolates the object/composition-law distinction between Smithe's programme and ours. Section 5 states what the convergence buys a grant that proposes to found a theory of compositional agents.

2. Preliminaries: the object and its dictionary

We recall only what the four fronts use. Proofs of the cited facts live in their sources and in the author's expository notes; none are reproduced here.

Containers and directed containers. A *container* is a pair $(S \triangleleft P)$ of a set S of *shapes* and a family $P \colon S \to \mathbf{Set}$ of *positions*; it presents the polynomial endofunctor $X \mapsto \sum_{s \in S} X^{P(s)}$ on $\mathbf{Set}$. A *directed container* equips $(S \triangleleft P)$ with operations $(o, \downarrow, \oplus)$ — a root position, a re-indexing, and a position addition — satisfying five laws (D1–D5) which are exactly the conditions making the induced functor a comonad [1]. Write $\mathbf{DCont}$ for the category of directed containers; the positions of a directed container form a $\triangleleft$-comonoid in the category **Poly** of polynomial functors under substitution [3].

Theorem 1 (Ahman–Uustalu [1])**.** [EXTERNAL] *There is an equivalence* $\mathbf{DCont} \cong \mathbf{Cat}$ *between directed containers and small categories. A directed container's shapes are the objects, its positions over a shape are the morphisms out of that object, the root is the identity, and* $\oplus$ *is composition. Under this equivalence, directed-container morphisms are the* cofunctors *(equivalently retrofunctors), so* $\mathbf{DCont} \cong \mathbf{Cof}$ *as well.*

Theorem 1 is the object of this entire note. Three of its instances recur below and deserve names in the polynomial language.

- **Delooping.** A monoid T is a one-object category BT; as a comonoid this is the representable y^T [4, Prop. 2.7].
- **Codiscrete comonoid.** For a set S, the monomial Sy^S carries a canonical comonoid structure — the codiscrete groupoid on S, equivalently the store comonad. We have elsewhere written this ΔS and shown $\Delta S \otimes \Delta T \cong \Delta(S \times T)$ [LEAN].
- **The fibrewise op.** The operation that reverses a container's positions fibrewise is what makes a container *directed*, and it is the same operation that reads a monad lifting on **Cont** as a comonad — hence a category — on **Set**. It is the hinge of front 3.1.

Agents are directed containers. The reading that turns Theorem 1 into an applied statement is due to the Poly/interaction school and, in the agent setting, to the framing used at Kodamai: an agent is a container whose shapes are the questions it can be asked and whose positions are the answers it can give [3]. A morphism to the representable y is a single oracle (model) call; a container morphism is a delegation; a comonad structure is the harness that lets an agent run. The directedness is not decoration — it is precisely the composability of an agent's own replies, and by Theorem 1 it is exactly a category structure on the agent's roles. We take this reading as given and study its consequences; it is the shared premise of every front below.

3. Four fronts on one theorem

3.1. **Classified into it: effectful agents are categories one level down.** An agent that carries an effect is modelled not by a bare container but by a *lifting* of a monad along the container fibration — a proof-relevant refinement of a predicate transformer. The natural question is which liftings exist, and the answer turns out to be another instance of Theorem 1.

Theorem 2 (Reader and State liftings are categories; author [PROVED] [LEAN])**.** *The proof-relevant polynomial monad liftings of the Reader monad* y^E *are, one level down, exactly* E*-indexed small categories; the sound liftings of the State monad embed the product category* $\mathbb{S} \times \mathcal{C}$*, and the machine-checked instances count the small monoids on their fibres. In each case a lifting on* **Cont** *is, via the fibrewise op, a comonad on* **Set** *— that is, a category.*

The point for this note is not the classification's internal detail but its shape: the objects that classify effectful agents *are the equivalence itself*, relocated one categorical level down. The equivalence $\mathbf{DCont} \cong \mathbf{Cat}$ is not merely the model of a bare agent; it recurs as the answer to "which effectful agents are there?" The hinge is the fibrewise op of Section 2, which is simultaneously what makes a container directed and what reads a lifting as a category. *The same operation that gives us the object gives us its refinement.*

Remark 3 (Scope, honestly)**.** The Reader classification and the sound half of the State classification are proved and, for the witnessing instances, Lean-verified. The *completeness* of the State classification (that every sound lifting is of the exhibited form) rests on a single holonomy-triviality lemma that is argued but not yet fully formalised; it is **[open]** at the level of a machine-checked converse. Nothing in this note depends on that converse: front 3.1 needs only that the equivalence is the classification *target*, which the proved direction already establishes.

3.2. **Cited as infrastructure: an external programme reaches for the same object.** The second front is the one that turns a self-assessment into external evidence. Smithe's *Compositional Active Inference II* [4] builds a categorical semantics for predictive-coding agents. Its substrate is Spivak-style polynomial dynamics: an open dynamical system is a map out of the codiscrete comonoid Sy^S on a state space (Proposition 3.2), and the condition for a coalgebra to be such a system is stated as its being a $\triangleleft$-comonoid homomorphism. To license this substrate the paper invokes Theorem 1 directly, at Proposition 2.7, in words worth quoting in full:

> "Famously, $\triangleleft$-comonoids correspond to categories and their comonoid homomorphisms are cofunctors [8]. If T is a monoid, then the comonoid structure on y^T ... witnesses it as the category BT. Monomials of the form Sy^S can be equipped with a canonical comonoid structure witnessing the codiscrete groupoid on S." [4, Prop. 2.7]

Reference [8] there is Ahman–Uustalu. The word "Famously" does the sociological work: the equivalence is invoked as settled background, neither proved nor re-derived. Every named instance in that passage — $\triangleleft$-comonoids as categories, comonoid homs as cofunctors, y^T as BT, Sy^S as the codiscrete groupoid — is an instance we also use.

Proposition 4 (Independent recognition; [EXTERNAL], deep-read verified)**.** *An external programme (Smithe, Topos Institute / VERSES; active-inference community) rooted in a tradition disjoint from container theory uses* $\mathbf{DCont} \cong \mathbf{Cat}$ *as settled infrastructure for the dynamical-systems substrate of compositional agents. The convergence is on the* object*: two independent programmes model a composable agent by the same equivalence.*

Two caveats keep the proposition honest, and both are load-bearing for the contrast of Section 4. First, in [4] the equivalence underwrites the *dynamical-systems substrate* (the state comonoids Sy^S), not the mechanism by which agents are composed — there, agents compose as Bayesian lenses, discussed next. Second, the citation is verified against a deep read of the source: reference [8] at Proposition 2.7, verbatim as quoted. We therefore lean on it; we do not paraphrase a summary of it.

3.3. **Generalized upward: the topos refinement.** The third front approaches Theorem 1 from above. The equivalence identifies a directed container with a category via a *comonad* on **Set**; asking which comonads on **Set** arise, and pushing the answer past the polynomial case, leads into topos theory. Pullback-preserving comonads on **Set** correspond to Grothendieck toposes with enough points, via skyscraper/stalk constructions — a topos-theoretic generalization of the situation $\mathbf{DCont} \cong \mathbf{Cat}$ describes. The community reaches the theorem from the toposes-with-enough-points end; the container programme reaches it from the directed-container end; the target is shared. The most developed written treatment of comonads-as-spaces adjacent to this circle of ideas is Fairbanks–Carlson–Spivak, *Comonads as Spaces* [5], which situates comonads (directed containers among them) in an explicitly spatial setting.

Remark 5 (Signpost, not a load-bearing citation)**.** The precise topos correspondence (pullback-preserving comonads $\iff$ Grothendieck toposes with enough points) is community knowledge whose cleanest reference — Garner's "Ionads" and the MathOverflow thread that assembles it — the author has not yet deep-read. We therefore present front 3.3 as a [SIGNPOST]: the generalization exists and is worth stating as evidence of the theorem's reach, but it is named, not leaned on. The citable anchor here is [5], which is deep-read; the sharper topos statement awaits a promotion pass.

3.4. **Bounded from outside: two witnesses at the edge.** The fourth front charts where the equivalence *stops*, which is as much a part of knowing an object as knowing where it holds. A directed container is a *polynomial* comonad; some functors that look categorical are not polynomial, and fail to be categories for a common, diagnosable reason. Two witnesses, one on each side of the monad/comonad duality, make the boundary two-sided.

Proposition 6 (Two boundary witnesses; author [PROVED] / [COMPUTED])**.** *On the monad side, the bag (finite-multiset) functor is not a container: a direct count gives* $|\mathrm{Bag}(2 \times 2)| = 10 \neq 9$*, breaking the product-preservation a polynomial functor would satisfy. On the comonad side, the multigraph comonad* $F(X) = (X^3 + X^2)/2 + X$ *— the coequalizer of the identity and the swap on the quiver comonad* $X^3 + X$ *— is not polynomial. Both fail the same test: neither is the kernel pair of its terminal map* $X \to 1$*, the condition characterising the polynomial (hence categorical) case.*

The bag count is the author's computation; the multigraph comonad is verified verbatim against its source and belongs to the same spatial circle as front 3.3 [5]. The moral is that "a composable agent is a category" is a *substantive* claim with a charted exterior: the quotients that identify distinct reply-histories (the swap on X^3, the multiset forgetting of order) leave the polynomial world, and with it the guarantee that composition is associative and unit-respecting on the nose. An agent whose interface secretly quotients its histories is not, in this precise sense, a category — and front 3.4 says exactly which quotients do the damage.

4. Object-level convergence, composition-level divergence

Fronts 3.1–3.4 establish that **DCont** $\cong$ **Cat** is the object. It remains to say clearly how the two programmes that share this object nonetheless *differ*, because the difference is exactly where each programme's own contribution lives, and conflating them would misrepresent both.

Smithe: composition as lens-wiring in a bicategory. In [4] agents are open dynamical systems whose interfaces are Bayesian lenses, and they compose in monoidal bicategories of "hierarchical inference systems" (Definition 3.15; its differential version, Definition 3.21). The systems that live over an external-hom polynomial and "control optics" are named *cilia* (Definition 3.8, Remarks 3.9–3.10). Wiring uses Bayesian-lens composition — a forward map paired with a state-dependent backward inversion — together with a distributive law of the internal hom over the tensor (Definition 3.13) used to assemble the bicategory. Composition, in short, is lens-wiring, and its coherence is bicategorical.

Ours: composition as a Zappa–Szép weld obstructed in H^2. Our composition law is different in kind. Two agents that share a resource each present their own handoff category, $\mathcal{C}$ and $\mathcal{D}$, and running them together is not a functor along one topology but an *interleaving* of two. The joint system is a Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$ — equivalently a matched-pair distributive law $\lambda\colon \mathcal{DC} \Rightarrow \mathcal{CD}$ of one agent category over the other. That a distributive law of directed containers assembles this weld, generalizing the monoid Zappa–Szép product, is Ahman–Uustalu's [2]; what is the author's is the applied reading and the obstruction. Unlike a functor, such a weld can fail to exist, and the author has proved that the failure is measured in a single cohomological degree: under the standard hypothesis, the weld exists iff a class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ vanishes, with re-entrancy the canonical nonzero class ($[\omega] = \varepsilon$, Lean-verified). This is a *monad* distributive law welding two categories; the obstruction is cohomological.

The precise relationship. We state the relationship as carefully as the evidence allows.

Proposition 7 (Convergent peers; contrast, do not conflate)**.** *Smithe's programme and the author's converge on the object — a composable agent is a directed container, i.e. a small category (Theorem 1, Proposition 4) — and diverge on the composition law:*

(i) [4] *composes agents as Bayesian lenses in a monoidal bicategory of cilia;*

(ii) *the author composes agent* categories *by the Zappa–Szép weld $\mathcal{C} \bowtie \mathcal{D}$ of Ahman–Uustalu* [2]*, adding an obstruction in H^2.*

These are structurally different mechanisms.

Two points of precision guard against the two ways this could be overstated. First, [4] is *not* distributive-law-free: it contains a distributive law (Definition 3.13). But that is a distributive law of an internal hom over a tensor, used to build a bicategory — *not* a monad distributive law welding two agent categories in the Beck / Zappa–Szép sense. When we say our composition is a "distributive-law weld" we mean the latter; the two gadgets share a name and nothing else. Second, and correspondingly, we do *not* claim that Smithe uses the weld, nor that anyone but the author composes agents by $\mathcal{C} \bowtie \mathcal{D}$. The claim of this note is the convergence on the *object*. The divergence on the composition law is not a defect in the convergence; it is the map of where each programme's own frontier lies.

5. What the convergence buys

A programme that proposes to found a theory of compositional agents must first pin down what an agent *is*, and then earn confidence that the choice is not idiosyncratic. This note reports that the choice — **DCont** $\cong$ **Cat** — is load-bearing four times over, and that one of the four loads is external.

- **Foundation, not fiat.** The object is not chosen for convenience: effectful agents *classify into* it (3.1), its exterior is *charted* (3.4), and it *generalizes upward* into topos theory (3.3). An object with a proved classification, a charted boundary, and a generalization frontier is a foundation.
- **External validation.** An independent group in a disjoint tradition reaches for the same equivalence and calls it "famous" (3.2). Convergence from the active-inference end is the strongest available evidence that this is the field's object, not merely ours.
- **A clean division of labour with the frontier.** The object-level question is settled and shared; the live research is the *composition law*. There the programmes genuinely part: lens-wiring for predictive coding, a cohomologically obstructed weld for shared-resource orchestration (4). The grant's own contribution — the weld $\mathcal{C} \bowtie \mathcal{D}$ and its class $[\omega] \in H^2$ — is precisely located as the composition law over an object the whole field already agrees on.

The hinge across all of this is the fibrewise op: the operation that makes a container directed is the operation that reads an effectful lifting as a category and, dually, reads a monad as a comonad. It is why the same equivalence can be a bare-agent model, an effectful-agent classification, and a topos special case at once. *Fair is foul, and foul is fair*: the op that looks like a bookkeeping reversal is the structure doing the work. That one theorem should be a target, an import, a frontier, and a boundary — and that an outside programme should reach for it independently — is the operational meaning of having chosen the right object.

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References

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[2] D. Ahman, T. Uustalu. *Distributive laws of directed containers.* Progress in Informatics, No. 10 (2013), 3–18, DOI 10.2201/NIIPI.2013.10.2. (Source of the weld: a distributive law of directed containers assembles the joint container, generalizing the monoid Zappa–Szép product.)

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[5] A. D. Fairbanks, K. Carlson, D. I. Spivak. *Comonads as Spaces.* arXiv:2607.15091, 2026. (Spatial reading of comonads; source of the multigraph-comonad boundary witness.)

[6] MacBeth. *The pairwise Zappa–Szép criterion.* Kodamai note, 10 June 2026 (proved).

[7] MacBeth. *The global-closure obstruction* (G) *is a cohomology class.* Kodamai note, 11 June 2026 (proved).

[8] MacBeth. *The re-entrancy obstruction is the token-mutation bit.* Kodamai note, 20 July 2026 (proved; $[\omega] = \varepsilon$ core Lean-verified, `Reentrancy.lean`).

[9] MacBeth. *Reader and State proof-relevant monad liftings are small categories.* Kodamai proof + Lean development, 9–11 August 2026 (Reader and sound-State proved; witnessing instances Lean-verified; State completeness open).

Kodamai